

Representation and Communication of Digital Information II

Laboratórios de Sistemas e Serviços

Mário Antunes

Universidade de Aveiro

2026

Introduction

As an engineer, writing code is only half of the job.

- The other half is **communicating** what that code does.
- Audience: Other developers, managers, clients, or your future self.
- Medium: Documentation, reports, theses, or presentations.
- Objective: Use tools that allow for versioning, automation, and professional quality.

The Cost of Poor Documentation I

What happens when we don't document?

- **Wasted Time:** Developers spending hours trying to understand a function.
- **Errors:** Misusing an API because the documentation was unclear.
- **Onboarding:** New team members taking months instead of weeks to be productive.

The Cost of Poor Documentation II

- **Maintenance:** Returning to your own code after 6 months and not knowing why you did something.
- **Technical Debt:** Unclear code/docs lead to hacks and workarounds.
- **Silos:** Knowledge is trapped in one person's head.

Markdown

What is Markdown?

A lightweight markup language with plain-text-formatting syntax.

- **Origin:** Created by John Gruber in 2004.
- **Philosophy:** “Readability, above all else.”
- **Format:** Plain text that can be converted to many formats (HTML, PDF, DOCX).
- **Usage:** GitHub (READMEs), Documentation (MkDocs, Sphinx), Academic notes.

Markdown is not a single standard, but has “flavors”:

- **CommonMark:** The attempt to create a highly compatible standard.
- **GitHub Flavored Markdown (GFM):** Adds tables, task lists, and autolinks.
- **Pandoc's Markdown:** The most powerful, adding citations, footnotes, and metadata.

Use the # symbol:

- # Title (H1)
- ## Section (H2)
- ### Subsection (H3)
- #### Sub-subsection (H4)

Markdown Syntax: Emphasis

- `*italic*` or `_italic_`
- `**bold**` or `__bold__`
- `***bold italic***`
- `~~strikethrough~~`

Unordered Lists: * Item 1 * Item 2 * Sub-item 2.1 * Sub-item
2.2

Ordered Lists: 1. First step 2. Second step 3. Third step

Markdown Syntax: Links and Images

- **Link:** `[Link Text](URL)`
 - `[University of Aveiro](https://www.ua.pt)`
- **Image:** `![Alt Text](Path/to/image)`
 - `![Course Logo](../../assets/logo.svg)`

Markdown Syntax: Code

- **Inline:** Surround with backticks: ``print("Hello")``.
- **Blocks:** Use triple backticks with language identifier.

```
python def hello(): print("Hello World")
```

Markdown Syntax: Tables and Tasks (GFM)

Task	Priority	Status
Plan	High	Done
Write	Medium	Active

- **Task Lists:**

- ☒ Complete Class 09
- ☒ Research Class 10
- ☐ Write Class 10 slides

Diagrams in Markdown: Mermaid.js I

Many platforms (GitHub, GitLab, Obsidian) support **Mermaid** for diagrams.

```
graph TD
  A[Start] --> B{Is it data?}
  B -- Yes --> C[Process]
  B -- No --> D[End]
```

Diagrams in Markdown: Mermaid.js II

Mermaid supports many diagram types: * **Flowcharts** *
Sequence Diagrams * **Gantt Charts** * **Class Diagrams** *
Entity Relationship Diagrams

It allows you to keep your diagrams **version-controlled** alongside your code.

LaTeX

What is LaTeX?

A high-quality typesetting system for technical and scientific documentation.

- **Focus:** Separating content from style.
- **Strength:** Mathematical notation, citations, and large documents.
- **Engine:** Based on TeX, created by Donald Knuth.
- **Standard:** Used for MSc Theses, PhD Dissertations, and Scientific Papers.

```
\documentclass{article}
\usepackage[utf8]{inputenc}

\title{My Report}
\author{Mário Antunes}

\begin{document}
\maketitle

\section{Introduction}
Hello world!
\end{document}
```

The `\documentclass{...}` defines the overall layout:

- **article:** For short reports or papers.
- **report:** For longer documents with chapters (like an MSc Thesis).
- **book:** For full-length books.
- **beamer:** For creating presentation slides.

Beamer allows you to create PDF slides using LaTeX syntax.

- **Frames:** Each slide is a `\begin{frame} ... \end{frame}`.
- **Themes:** Professional, academic looks (e.g., Warsaw, Madrid).
- **Overlays:** Control when content appears (e.g., `\pause`).
- **Consistency:** The math and code in your slides will look identical to your paper/thesis.

Packages extend LaTeX's capabilities:

- **graphicx**: For including images (`\includegraphics`).
- **hyperref**: For clickable links and cross-references.
- **geometry**: For changing page margins.
- **amsmath**: For advanced mathematical symbols.
- **biblatex**: For managing bibliographies.

LaTeX is the industry standard for writing math.

- **Inline:** $E = mc^2$.
- **Display:**

$$\int_a^b f(x)dx$$

- **Fractions:** `\frac{numerator}{denominator}`
- **Sums:** `\sum_{i=0}^n x_i`
- **Greek Letters:** `\alpha`, `\beta`, `\gamma`, `\pi`
- **Sub/Superscripts:** `x_i`, `x^2`
- **Matrices:** `\begin{pmatrix} a & b \\ c & d \end{pmatrix}`

Managing references manually is a nightmare.

1. Create a .bib file with your references.

```
@article{einstein1905,  
  author = {Albert Einstein},  
  title = {On the Electrodynamics of Moving Bodies},  
  journal = {Annalen der Physik},  
  year = {1905}  
}
```

2. Cite it: Einstein showed that
`\cite{einstein1905}`...

Pandoc

The “Swiss-army knife” of Documents

Pandoc is a command-line tool to convert files from one markup format into another.

- **Input:** Markdown, LaTeX, HTML, DOCX, EPUB.
- **Output:** PDF, HTML, LaTeX, Slides (Beamer/RevealJS), DOCX.
- **Command:** `pandoc input.md -o output.pdf`

How Pandoc Works I

1. **Reader:** Parses the input (e.g., Markdown).
2. **Intermediate Representation:** Converts to an internal “AST” (Abstract Syntax Tree).
3. **Writer:** Generates the output (e.g., PDF via LaTeX).

- **YAML Metadata:** Use a `config.yml` or a block at the top of the `.md`.
- It controls titles, authors, dates, and template variables.
- **Templates:** You can provide your own LaTeX or HTML template to control every detail of the output.

Pandoc's power can be extended using **filters**.

- Filters are small scripts that modify the document's internal AST before it is written.
- **Types:** Python filters (`panflute`), Lua filters (built-in).
- **Usage:**
 - Automatically number figures.
 - Transform table formats.
 - Inject custom LaTeX for specific blocks.

Useful Documents

- **Format:** Keep it simple, searchable, and professional.
- **Tools:** Use LaTeX templates (e.g., `moderncv`) or Markdown + Pandoc.
- **Key Sections:** Education, Experience, Skills, Projects.
- **Pro Tip:** Keep your CV in a Git repository and generate multiple versions (e.g., detailed vs 1-page).

The MSc Thesis: Typical Structure

1. **Front Matter:** Title, Abstract, Acknowledgments, ToC.
2. **Introduction:** Motivation, Problem, Objectives, Contributions.
3. **State of the Art:** Literature review and comparison.
4. **Proposed Solution/Methodology:** Architecture, design, and implementation.
5. **Evaluation:** Experimental setup, results, and discussion.
6. **Conclusion:** Summary and Future Work.
7. **Back Matter:** Bibliography, Appendices.

The most important part of your thesis is your original contribution.

- It's not just “building a system”.
- It's “solving a problem” or “improving a process” using engineering principles.
- You must clearly state **what is new** in your work.

- **Executive Summary:** For busy managers.
- **Methodology:** How you did it.
- **Findings:** What you discovered.
- **Recommendations:** What should be done next.
- **Automation:** Use a Makefile to generate reports from Markdown.

- **Sphinx:** The industry standard for Python docs.
- **MkDocs:** Modern, fast, and uses Markdown.
- **Docstrings:** Write documentation inside your code.

Python Documentation II: Docstrings

```
def calculate_area(radius: float) -> float:  
    """Calculates the area of a circle.
```

Args:

radius: The radius of the circle (must be positive).

Returns:

The calculated area.

Raises:

ValueError: If radius is negative.

```
    """
```

```
if radius < 0:
```

```
    raise ValueError("Radius cannot be negative")
```

```
return 3.14159 * (radius ** 2)
```

Summary

- **Communication:** Is a core engineering skill.
- **Markdown:** For daily notes, READMEs, and quick reports.
- **LaTeX:** For formal academic and high-quality technical documents.
- **Pandoc:** To bridge all formats and automate your workflow.
- **Consistency:** Use professional templates and version control for everything.